

Milestone 3 — Data Preparation, Mining, and Visualization

Taya Yakovenko — AI's Impact on the Job Market

Data Preparation

The original Kaggle dataset was rejected after preliminary inspection showed near-zero correlations across all variables — a sign of synthetic data with no real signal. The analysis uses three real-world sources instead: Frey & Osborne automation probability scores (702 occupations), BLS employment projections 2024-2034, and the ILO 2025 GenAI Exposure Index (427 ISCO-08 occupations).

The F&O and BLS datasets were joined on SOC code, yielding **606 matched occupations** (86% match rate). The ILO data was bridged via the BLS ISCO-08 × SOC crosswalk, covering 600 of the 606 (99%).

Four features were engineered:

- **risk_tier** — Low / Medium / High automation risk bands
- **occupation_group** — 22 groups from SOC major code
- **adaptive_capacity_score** — normalized wage + education combined into a 0-1 resilience index
- **vulnerable** — binary flag: high automation risk + bottom-third adaptive capacity → **144 occupations (23.8%)**

Key data facts: automation probability is bimodal (most jobs cluster at the extremes, not the middle); 65.3% of occupations show positive projected employment growth despite widespread automation risk; the vulnerable group has a median wage of \$41,503 vs. \$72,293 for non-vulnerable occupations.

Data Mining

Two techniques were applied.

K-Means clustering (k=3) on automation risk and adaptive capacity features produced three interpretable segments:

Cluster	Share	Avg Automation	Avg Wage	Avg Growth
High Risk / Low Resilience	50.8%	85%	\$49,287	−1.75%
Low Risk / Stable	37.5%	27%	\$66,965	+2.58%
Low Risk / High Skill	11.7%	9%	\$126,322	+5.51%

Half the matched occupations are in the bottom cluster — high automation exposure with virtually no economic buffer.

Linear regression (target: projected employment change) confirmed both hypotheses:

- Automation risk negatively predicts employment change (coefficient −4.15; each +10pp automation → −0.42pp growth)
- But the full model — automation + wage + education + 21 occupation-group dummies — explains only **$R^2 = 0.175$** on the test set. Most of what determines a job's trajectory lies outside AI exposure.

The largest single predictor in the model was the Computer & Math sector dummy (+12.4pp), reflecting structural tech demand independent of automation risk.

Data Visualization

14 figures were produced across 6 chart types. The full list is in

`report/04_data_visualization.md`. Key visuals:

- **Scatter plot (fig2)** — shows the Automation Paradox directly: negative trend, but wide scatter means many high-risk jobs still grow
 - **Ranked bars (fig3, fig4)** — clerical/production jobs face the steepest declines; healthcare/tech lead growth
 - **Wage gap (fig5, fig10)** — \$30,650 gap between high and low risk tiers; vulnerable workers have almost no high-earners in their cohort
 - **Dumbbell chart (fig13)** — sector-level risk shift from 2013 to 2025: physical sectors de-risked, knowledge sectors newly exposed; Computer & Math reversed by +43 percentage points
 - **Scatter (fig12) and diverging bar (fig14)** — occupation-level flip: writers, analysts, mathematicians are newly exposed; brickmasons, cement masons, landscapers are de-risked
-

This document was drafted with the assistance of Claude (Anthropic) as a writing and analysis tool. All data, findings, code, and analytical decisions are the author's own.